

SYDNEY BADGERYS CREEK, Australia

WMO: 947520

Lat: 33.897S

Lon: 150.728E

Elev: 82

StdP: 100.34

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.6	2.8	-2.3	3.1	14.5	-1.1	3.5	12.8	11.0	16.6	9.5	15.6	1.2	230	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	35.6	21.0	33.0	20.6	30.7	20.1	23.5	30.1	22.6	28.5	21.9	27.4	4.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	16.8	24.9	21.1	15.9	24.3	20.4	15.2	23.7	70.6	30.4	67.2	28.5	64.5	27.4	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	7.8	6.8	-0.8	40.8	0.8	2.5	-1.4	42.7	-1.9	44.1	-2.3	45.6	-2.9	47.4		
			-1.1	25.6	1.0	1.8	-1.8	26.9	-2.4	28.0	-2.9	29.1	-3.6	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	23.1	22.7	20.8	17.7	13.9	11.6	10.7	11.9	14.8	17.4	19.7	21.6
	DBStd	5.02	3.14	2.81	2.44	2.41	2.18	1.95	1.93	2.02	2.68	3.04	3.25	3.03
	HDD10.0	26	0	0	0	0	1	7	13	5	0	0	0	0
	HDD18.3	1017	2	1	6	40	137	202	236	201	111	56	21	6
	CDD10.0	2624	404	355	334	229	123	55	36	63	144	228	292	361
	CDD18.3	574	148	123	82	19	1	0	0	0	6	26	62	108
	CDH23.3	4783	1228	900	487	153	12	0	1	7	121	353	612	909
	CDH26.7	1882	564	359	134	34	1	0	0	0	30	124	254	382
Wind	WSAvg	2.8	2.9	2.8	2.6	2.5	2.4	2.6	2.6	3.0	3.0	3.0	3.0	3.0
Precipitation	PrecAvg	796	93	88	84	72	61	59	33	47	42	62	78	62
	PrecMax	1243	269	405	310	365	186	312	131	241	145	183	370	186
	PrecMin	372	9	8	3	0	0	0	0	0	0	0	7	0
	PrecStd	260	65	87	65	77	54	70	31	59	33	51	74	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	37.6	32.9	30.3	24.8	20.7	22.1	24.2	30.3	33.5	37.1	37.7
		MCWB	22.2	22.0	21.2	18.4	16.0	13.5	12.8	13.5	16.0	17.1	18.8	20.8
	2%	DB	35.5	33.8	29.8	27.0	22.8	18.8	19.2	21.5	26.7	30.0	32.8	33.9
		MCWB	21.7	22.2	21.0	17.7	15.3	13.2	11.9	12.9	15.1	17.1	19.6	20.4
	5%	DB	32.2	30.5	27.7	24.9	21.1	17.6	17.6	19.6	24.0	27.3	29.2	30.7
		MCWB	21.4	21.6	20.6	17.6	14.7	12.4	11.3	12.0	14.6	16.9	18.9	19.5
	10%	DB	29.5	28.3	25.9	23.2	19.7	16.5	16.3	18.0	21.7	24.5	26.7	28.3
		MCWB	20.8	20.9	19.7	17.0	13.9	12.0	10.7	11.2	13.9	16.1	18.2	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	25.6	24.3	20.6	18.6	15.9	15.7	15.6	18.3	20.4	22.6	23.2
		MCDB	33.3	33.1	27.7	26.4	20.9	17.6	18.2	20.5	23.7	26.9	29.8	30.6
	2%	WB	23.3	23.6	22.7	19.5	16.9	14.6	13.7	14.1	17.0	19.0	21.2	22.2
		MCDB	31.1	29.7	26.3	23.7	20.9	16.7	16.3	18.9	23.0	25.1	27.9	29.3
	5%	WB	22.4	22.6	21.6	18.6	15.7	13.8	12.5	13.1	16.0	18.1	20.2	21.3
		MCDB	29.2	27.7	25.7	22.5	19.7	16.1	15.7	17.5	21.3	23.9	26.4	27.6
	10%	WB	21.7	21.8	20.7	17.9	14.8	13.1	11.5	12.2	15.0	17.1	19.2	20.5
		MCDB	27.4	26.4	24.4	21.6	18.5	15.4	15.0	16.5	20.0	22.7	24.7	26.1
Mean Daily Temperature Range	5% DB	MDBR	11.7	10.6	10.1	11.4	12.2	10.6	12.2	13.1	13.5	12.9	11.8	11.7
		MCDBR	17.9	15.6	13.6	14.3	14.4	12.6	14.7	16.3	18.1	18.8	17.6	17.2
		MCWBR	5.7	5.6	5.6	6.2	7.5	7.3	8.2	8.4	7.8	7.3	6.4	5.8
	5% WB	MCDBR	14.1	12.7	11.2	11.7	12.2	9.7	11.7	13.7	14.9	14.4	14.1	13.8
		MCWBR	5.2	5.3	5.2	5.7	7.0	6.4	7.4	7.7	7.2	6.4	5.8	5.5
Clear-Sky Solar Irradiance	taub	0.394	0.390	0.361	0.332	0.315	0.305	0.296	0.303	0.335	0.357	0.383	0.394	
	taud	2.432	2.463	2.523	2.577	2.573	2.601	2.609	2.581	2.490	2.459	2.429	2.408	
	Ebn at Noon	944	926	918	892	855	839	870	911	929	948	948	947	
	Edh at Noon	122	114	100	85	75	69	71	82	102	113	122	126	
All-Sky Solar Radiation	RadAvg	6.41	5.45	4.68	3.81	3.04	2.40	2.81	3.71	4.93	5.70	6.01	6.59	
	RadStd	0.50	0.51	0.42	0.33	0.21	0.21	0.21	0.23	0.30	0.55	0.56	0.61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	-2	-85	+184

Nomenclature: See separate page